Logic as a Consequence of the Ontological Self-Consistency of the Universe

W-Structure, Local Realization, and the Mathematical Language of Description

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Revision 4 — 9 November 2026. A complete revision of “Logic as a Consequence of the Non-Contradiction of the Universe,” revision 2, incorporating the author’s clarifications of 9 November and a renewed review of revision 3.

Basis of the text. The author’s concept of W-structure and the explanations supplied by the author. The mathematical notation below describes relations; it does not prescribe a mode of existence to the Universe. External results are used only where properties of a formal language are explicitly being considered; they do not replace the justification of ontological claims.

Abstract

In W-structure, self-consistency does not arise through choice, a search through alternatives, or the elimination of contradictions. A non-self-consistent actualization is not another state of the Universe: it is not realized as a state. Accordingly, change from a system’s beginning to its W is not a transition from contradiction to non-contradiction. Self-consistency pertains to every realized change and, simultaneously, to the whole in its entirety.

On this understanding, logic does not govern existence from outside. Logical laws express the necessary relations of what exists. The purpose of this article is to explain this direction of grounding without equating it with the claim that any chosen formal language has already been derived from the symbol `C=0` alone.

Three subjects are distinguished: ontological actualization, its mathematical representation, and the knowledge available to a subsystem. The unique consistent history is not replaced by a global queue of events. A local, nonzero, indivisible act of change gives rise to its own derived local tick of time. The chain `00—0W≡10—1W≡20—…—W0—WW≡00` describes relations between levels, not a universal counter.

For a finite event record, a language of statements about realization is constructed. The meanings of “and,” “or,” persistent exclusion, and persistent implication are obtained from explicitly stated semantic requirements on regions of realization. The laws are then proved directly from these meanings. This retains the mathematical core of the original article without presenting the adopted semantics as an ontological mechanism.

The proper scope of Gödel’s theorems is established separately: the Universe is not automatically a formal calculus, but an algorithmic description of it is not exempt from mathematical theorems merely because it describes W.

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1. Self-Consistency Is a Condition of Existence, Not a Result to Be Achieved

1.1. The ontological meaning

In this conception, self-consistency does not mean the compatibility of several independently arising descriptions. It means that a local state or change cannot exist outside the consistent actualization of the whole.

The following misrepresents this meaning:

> First, individually admissible alternatives arise; the Universe then compares them and selects a non-contradictory one.

Such a formulation introduces existing alternatives and a selection procedure that are absent from the author's conception. In what follows, when ontology is at issue, the word "admissible" does not denote the output of an external checker.

Change in the Universe is already an actualization of self-consistency, not a search for it. The notation

$$C_{\text{local}} = C_{\text{global}} = 0$$

expresses the absence of realized contradiction both locally and in the whole. It does not specify a numerical function that the Universe computes and minimizes.

"The only possible state" does not mean a single motionless configuration. What remains unchanged is the condition of existence; configurations change. In particular,

$$0_S \longrightarrow W_S$$

does not mean

$$C(S) > 0 \longrightarrow C(S) = 0.$$

1.2. In what sense logic is a consequence

If the realization of relation A necessarily presupposes relation B, a logical expression such as `A follows from …` or `A ⇒ B` expresses the corresponding dependence. It does not compel the Universe to check A and then choose B. Realizing A without the B that it requires is not another realizable alternative of the same structure.

The direction of grounding is:

$$\boxed{\text{ontological self-consistency} \longrightarrow \text{necessary relations of what exists} \longrightarrow \text{logical notation for these relations.}}$$

The arrows in this diagram indicate a relation of grounding, not a temporal sequence. There is no initial era "without logic" followed by the emergence of logic.

1.3. Three levels that must not be conflated

Subject	What belongs to it	What does not automatically follow
Ontological actualization	Realized distinctions and relations of a single whole	That a subsystem knows or can prove all these relations
Mathematical description	Notation, formulas, models, and proofs	That every model written down is realized in the Universe
A subsystem's knowledge	Available observations, records, grounds, and errors	That a subsystem's lack of knowledge creates an ontological alternative

The existence of a written expression "A and not-A" does not imply the existence of the contradictory state it describes. The inscription itself is an actualized object; its content may fail to correspond to what it asserts.

This is precisely the distinction from problems formulated by intelligent systems. Such formulations may contain incompatible requirements, incomplete definitions, or errors. Correcting such a formulation is not the Universe correcting its own existence.

1.4. The status of grounds and definitions

This article retains the ontological content in the author's sense; it does not replace it with an external axiomatic framework. A rigorous text must, however, identify the grounds of each inference. Accordingly, the following are distinguished:

- **the content of the author's conception** — what exactly is asserted about the Universe;
- **the meaning of the notation** — what the introduced symbols denote;
- **a derivation** — what actually follows when those meanings are preserved;
- **an unresolved representational step** — where it has not yet been shown that the chosen notation preserves the required ontological property.

Defining a symbol's meaning does not add a law of the Universe. Yet calling an additional requirement a "definition" is not enough to make it a derived result. For example, the persistence of the truth of every statement does not follow merely from the persistence of the fact that an event has already occurred. This distinction will be maintained below.

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2. Indivisible Change, Derived Time, and the Chain of Levels

2.1. The nonzero atom of change

In the author's understanding, an elementary act is a nonzero, indivisible change: the realization of one atom of distinction. One such act gives rise to one derived, nonzero, minimal tick of time for the corresponding system.

These are not two independently given entities:

$$\boxed{\text{an elementary realized act} \quad \leftrightarrow \quad \text{its derived local tick.}}$$

Time is not an external container within which the act occurs. It therefore cannot, without justification, be represented as a jump taking zero time. Nor can an indivisible act be decomposed into smaller, actually realized acts while simultaneously retaining the claim that it is minimal.

For a local representation, the sequence of states of a system S may be denoted by

$$s_0^{(S)} \longrightarrow s_1^{(S)} \longrightarrow \cdots .$$

Here the index counts changes in S. If another system is assigned the same index, the numerical coincidence does not by itself assert physical simultaneity. Counting acts does not yet provide a conversion into seconds or a universal relation between the rates of different systems.

2.2. A point is not nothing

A point, in the author's sense used here, is a nonzero, indivisible structural atom. Its indivisibility does not mean its absence. Zero dimensionality, zero numerical measure in some external language, and the absence of a realized distinction are different claims.

This article does not assign a positive diameter or volume to a point: that metric meaning has not been obtained from the definition of an indivisible atom. Nor is `00` replaced by a point: according to the author's clarifications, `00` is not a pre-existing spatial point.

2.3. Uniqueness without a global queue

In this conception, each local system actualizes its own unique, non-contradictory trajectory, while the local trajectories jointly belong to a single self-consistent whole. They do not first become independent and then undergo synchronization.

"Global" denotes the scope of self-consistency, not a shared moment at which it is checked. Neither universal clocks nor a queue for updating all subsystems is introduced on this basis.

Two events not related by precedence may be listed in different orders. A difference between lists does not establish a difference between realized histories. It is a difference between mathematical representations unless a significant relation changed by the permutation is exhibited.

At the same time, discreteness alone does not prove that assigning a common counter mathematically is impossible. Such a counter can be added to a description. The question is whether it expresses a relation of the structure itself or is an arbitrary external labeling. The author's absence of a global "now" concerns the former issue, not a prohibition on writing numbers next to events.

Accordingly, this article does not use "a discrete step by itself rules out a global now" as a separately proved theorem. The more precise content is retained: derived local ticks do not by themselves provide a physically privileged global synchronization. A stronger prohibition requires an explicit derivation from the relations of the whole.

2.4. Structural closure of the levels

The general notation is:

$$00 \longrightarrow 0W \equiv 10 \longrightarrow 1W \equiv 20 \longrightarrow 2W \equiv 30 \longrightarrow 3W \equiv 40 \longrightarrow \cdots \longrightarrow W0 \longrightarrow WW \equiv 00.$$

Here `qW ≡ (q+1)0` denotes one completed whole in two interlevel relations: as the W of the preceding level and as the initial atom of the next. The sign `≡` does not denote an additional transition between two already distinct objects.

The chain is not a common calendar. `WW ≡ 00` does not, by default, denote a return to a previous global moment. Without separate justification, structural closure cannot be interpreted either as an infinite repetition of temporal cycles or as a single temporal passage followed by a halt.

Self-closure is not a logically circular proof. The validity of a proof is a separate question, considered in Section 10.

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3. A Mathematical Representation of One Consistent History

3.1. Why an event record is needed

Mathematics is used as a language. A description must therefore distinguish what it represents from what it adds for notational convenience. Neither sets, partial orders, nor lattices in the following sections are identified with matter or with a fundamental algorithm of the Universe.

We consider a finite event record of one level or of a selected finite domain. Having a complete history specified for a proof does not mean that an observing subsystem already possesses its complete description. In general mathematical statements, the record may have an arbitrary finite size. This is not a claim about a realized infinite set of systems.

Let E denote the events in this record. Distinct events are distinct actualized acts, not necessarily distinct configuration forms. Repetition of a form does not identify two acts of change with one another.

The notation `d ⩽ e` means that the realization of e in the history under consideration presupposes that d has already been realized. A partial order is used for the mathematical model. This relation serves as event precedence, not as an arbitrary relation between any two systems.

It is not claimed to have been proved that one such order can represent all necessary global consistency. If some relations of the history cannot be expressed as precedence between events, they must not be silently discarded. The following theorems concern a representation in which the relations being used are explicitly preserved.

3.2. A cut is not a global “now”

A *cut* is a down-set of events:

$$\mathcal{C} = J(E) = \{c \subseteq E : \ e \in c,\ d \preceq e \Rightarrow d \in c\}.$$

We write `c ≤ u` for a continuation of the record when `c ⊆ u`. This means inclusion of the events already represented, not an instruction for the Universe to move from one universal snapshot to another.

All cuts are elements of the mathematical representation; it is not asserted that each was a physically distinguished state of the entire Universe. Transitions between cuts allow reasoning about the compatibility of fragments of one history. They do not depict a choice between alternative ontological worlds.

To avoid conflating different meanings of zero, we use

$$c_{\varnothing} = \varnothing, \qquad T_E = E.$$

`c_∅` is a record with no events included, but with its domain and relations specified. It is not absolute nothingness, is not identical to `00`, and does not negate the nonzero character of the atom. `T_E` is the complete event cut of this record. Its relation to ontological W is discussed in Section 8.

3.3. Truth about realization, truth about the whole history, and knowledge

The elementary statement `p_d` reads: “event d is realized in the represented fragment.” Its region of realization is denoted by

$$U_d = \{c \in \mathcal{C} : d \in c\}.$$

If `c ⊆ u`, an event included in c remains included in u. This is persistence of the fact of realization. It **does not mean** that every current physical property of a configuration is unchanging or that every subsystem forever retains an accessible record of the event.

The following must be distinguished:

1. “d has already been realized in the fragment under consideration”;
2. “d belongs to the unique complete history”;
3. “the subsystem knows or has proved that d is realized.”

The first depends on the cut. The second does not turn a local act that has not yet occurred into one that has already occurred. The third requires a separate account of knowledge and evidence.

In particular, `c ⊢/ A` below denotes only the absence of realization of A in this semantics. It is not a statement that an algorithmic problem is undecidable, nor an acknowledgment of an ontological choice between A and not-A.

3.4. Persistent regions of realization

Call a region U *persistent* if

$$c \in U, \quad c \le u \quad \Rightarrow \quad u \in U.$$

Denote all such regions by $\mathscr{U}(\mathscr{C})$. They describe statements about what has been realized that retain their chosen historical meaning as the record is extended. They need not describe every kind of statement about the Universe.

For a finite, complete event record, every such U can be expressed by a formula of events:

$$\Phi_U = \bigvee_{c \in U} \bigwedge_{d \in c} p_d.$$

The empty conjunction denotes the condition `⊤`, which is always realized in the language; the empty disjunction denotes `⊥`. The formula is true at u if and only if some `c∈U` is included in u. By persistence, this is equivalent to `u∈U`.

This is a proof of the language’s expressive power. It does not create events or assert that the system searches through its future cuts.

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4. Connectives from the Meaning of Relations, Not from an External Calculus

4.1. Joint realization and realization expressed as an alternative

Let U and V be the regions of realization of A and B.

The condition “both are realized” holds exactly where both original conditions hold:

$$[A \wedge B] = U \cap V.$$

The condition “at least one is realized” holds exactly where A is realized or B is realized:

$$[A \vee B] = U \cup V.$$

The intersection and union of persistent regions are persistent. Here “or” does not mean that the Universe chose between two alternative histories; it concerns the content of a formula on a single event record.

The meanings of ⊤ and ⊥ are:

$$[\top] = \mathcal{C}, \qquad [\bot] = \varnothing.$$

These are the meanings of conditions in the language, not the names of two physical substances or states.

4.2. Persistent implication: a precise semantic requirement

We need to express a condition such that the joint realization of this condition and A admits no continuation without B. The condition itself must persist as the record is extended.

Among the persistent regions Z satisfying

$$Z \cap U \subseteq V,$$

we require the **greatest** one: it expresses all cases of such persistent dependence, rather than an arbitrarily selected subset of them. Denote it by $U \Rightarrow V$.

Requiring the greatest region specifies the meaning of “implies” in this language. It is not an additional law of the Universe. Choosing a smaller region would yield a different, narrower statement. A requirement of maximal expressive scope cannot be silently obtained from the prohibition of contradiction alone; it is made explicit here.

Theorem 1. Such a region exists, is unique, and is equal to

$$\boxed{U \Rightarrow V = \{c \in \mathcal{C} :\ \forall u \geq c\, (u \in U \Rightarrow u \in V)\}.}$$

Proof. Denote the right-hand side by H. If $c \in H$ and $c \leq v$, every continuation of v is a continuation of c. Hence $v \in H$: H is persistent. If $c \in H \cap U$, we may take $u = c$ in the definition, obtaining $c \in V$. Thus $H \cap U \subseteq V$.

Now let Z be any persistent region with $Z \cap U \subseteq V$, and let $c \in Z$. For every $u \geq c$, we have $u \in Z$. If, in addition, $u \in U$, then $u \in V$. Therefore $c \in H$ and $Z \subseteq H$. H is indeed the greatest such region. Two greatest regions are contained in one another and thus coincide. ∎

In particular,

$$Z \cap U \subseteq V \quad \Longleftrightarrow \quad Z \subseteq (U \Rightarrow V).$$

This is a proved relation between joint realization and implication, not an axiom imposed on the structure from above.

4.3. Exclusion is not the same as absence of realization

The condition “A is excluded” must be persistent and incompatible with the realization of A. The full expression of this meaning is the greatest persistent region disjoint from U:

$$\boxed{[\neg A] = U \Rightarrow \varnothing = \{c :\ \forall u \geq c\, (u \notin U)\}.}$$

Thus $c \notin U$ does not imply $c \in [\neg A]$. An event may not yet be included in c but may be present in a continuation of the record of the same unique history.

The terms must have fixed meanings. If negation denotes only the pointwise complement $\mathcal{C} \setminus U$, it will generally not be persistent. For example, “d has not yet occurred” may cease to hold after d. This is a legitimate statement of another kind, but it is not the same as “d is excluded.”

4.4. The resulting notation

Write $c \Vdash A$ when c belongs to the region of realization of A. The resulting conditions are:

$$
\begin{aligned}
c \Vdash p_d &\iff d \in c, \\
c \nVdash \bot, &\qquad c \Vdash \top, \\
c \Vdash A \wedge B &\iff c \Vdash A \text{ and } c \Vdash B, \\
c \Vdash A \vee B &\iff c \Vdash A \text{ or } c \Vdash B, \\
c \Vdash A \to B &\iff \forall u \geq c\, (u \Vdash A \Rightarrow u \Vdash B), \\
c \Vdash \neg A &\iff \forall u \geq c\, (u \nVdash A).
\end{aligned}
$$

This notation coincides with a known semantics of persistent propositions. However, the name of a calculus has not been used here to justify the choice: the meanings have been obtained from the stated semantic requirements on regions of realization.

Scope of the derivation: these operations have been shown to be determined by the specified meanings of the language. It has not been proved that the entire language, all its subject matter, and a complete ontology can be recovered from the uninterpreted symbol `C=0` alone. Nor is it claimed that other languages cannot describe other aspects of the same structure.

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5. Which Laws Follow from the Resulting Relations

5.1. Persistence of the language

Lemma 2. If `c ⊩ A` and `c ≤ u`, then `u ⊩ A`.

Proof. For events, this is preservation of inclusion. Intersection and union are persistent. For implication and negation, the cone of continuations of u is contained in the cone of continuations of c. Applying this reasoning according to the structure of the formula completes the proof. ∎

Persistence pertains to the selected statements about realization. It does not require the constancy of color, velocity, or any other current characteristic of a system.

5.2. Non-contradiction and an empty premise

Theorem 3. The following hold in this language:

$$
\neg(A \wedge \neg A), \qquad (A \wedge \neg A) \to B.
$$

Proof. If `c⊩¬A`, then A is not realized in any continuation of c, including c itself. The region of `A∧¬A` is therefore empty. Its negation holds everywhere. The implication from it to B also holds: there is no continuation that realizes the premise and violates the conclusion. ∎

This is a mathematical representation of the incompatibility between realization and its complete exclusion. It does not derive ontological self-consistency from a contradictory world assumed to exist beforehand.

The second formula is the so-called principle of explosion in this language of implication. It does not mean that a contradiction physically produces everything that exists or erases matter. If a description requires both A and its exclusion in one and the same respect, the description has no realization in this semantics. From the absence of suitable realizations, any condition follows formally; this is a property of descriptive entailment, not an action of the Universe.

5.3. Persistent dependencies

Theorem 4. The following schemata are obtained from Section 4:

Meaning	Formula
Preservation of a realized premise	`A → (B → A)`
Identity dependence	`A → A`
Application of a dependency	`(A ∧ (A → B)) → B`
Composition of dependencies	`((A → B) ∧ (B → C)) → (A → C)`
Extraction of a part of a joint realization	`(A ∧ B) → A`

Meaning	Formula
Inclusion in the realization of “at least one”	$A \to (A \vee B)$
Reasoning by two cases of realization	$((A \to C) \wedge (B \to C)) \to ((A \vee B) \to C)$
Transfer of exclusion along a dependency	$(A \to B) \to (\neg B \to \neg A)$
What is realized is not excluded	$A \to \neg\neg A$
Reduction of triple exclusion	$\neg\neg\neg A \to \neg A$

Proofs.

Preservation. If A is realized in u, it persists in every $v \geq u$. Hence the dependency $B \to A$ is realized in u, regardless of when B is realized.

Identity. Every continuation with A is a continuation with A. No separate procedure for selecting truth is required.

Application. If A and $A \to B$ are realized in u, the definition of the dependency applies to u itself; hence B is realized.

Composition. Suppose $A \to B$ and $B \to C$ are realized in u. In any $v \geq u$ where A is realized, B is realized, and then C. This is precisely what $A \to C$ means in u.

Part and union. These express the inclusions $U \cap V \subseteq U$ and $U \subseteq U \cup V$.

Cases. If every way of realizing A leads to C and every way of realizing B leads to C, then C is realized in any continuation realizing A or B.

Exclusion. Suppose the dependency $A \to B$ is realized. If some continuation excludes B, none of its further continuations can realize A: otherwise the dependency would entail the realization of B. Therefore A is excluded there.

Double negation. Once A is realized, no continuation can exclude A, since A persists. Hence $\neg\neg A$ is realized.

Triple negation. Suppose $\neg\neg\neg A$ is realized in u. If some continuation v realized A, the preceding result would give $\neg\neg A$ in v. But $\neg\neg\neg A$ excludes precisely $\neg\neg A$ in every continuation of u. Thus there is no continuation with A, and $\neg A$ is realized in u. ∎

These proofs do not use the list of laws of any calculus as an external ontological premise. The names of known logics are applied to the resulting descriptive apparatus later.

5.4. Compatibility cannot be reduced to pairwise checking

The three requirements p, q, and $\neg(p \wedge q)$ may be pairwise compatible but jointly incompatible. In a complete two-valued representation, the first pair holds at $p=q=true$, the second at $p=true, q=false$, and the third at $p=false, q=true$. No valuation satisfies all three simultaneously.

This example concerns constructed descriptions. It does not attribute a contradictory stage to the Universe. It shows why checking fragments of a description does not replace checking their joint realization.

5.5. What failure of a classical schema before the complete cut means

Theorem 5. For every proper cut $c \neq T_E$, there is a statement A such that neither $A \vee \neg A$ nor $\neg\neg A \to A$ is realized at c.

Proof. Choose $d \in E \setminus c$ and take $A = p_d$. The event is not realized at c. But the complete cut contains d and extends every continuation of c. Consequently, d is excluded in no continuation of c; that is, $\neg\neg p_d$ is realized at c. Both failures follow. ∎

The quantifiers matter: for each incomplete cut, there **exists** such a formula. It is not claimed that all formulas are undetermined before the complete cut. For example, $\top$, $A \to A$, and already realized events are determined earlier.

Nor does this mean that the Universe itself has not yet determined what is to happen. A statement about an event that has not yet occurred and a description of its place in the unique history have different meanings.

6. One History: Joint Compatibility and Forthcoming Realization

6.1. A common continuation of descriptions

Lemma 6. For any `u,v∈J(E)`, the union `u∪v` is a cut. The complete cut `T_E` extends every cut.

Proof. A predecessor of an event in the union lies in the down-set to which that event belongs. Hence it belongs to the union. ∎

This is a property of joint descriptions of one event history. It does not mean that the Universe first generated separate branches and subsequently merged them.

6.2. The weak law of excluded middle

Theorem 7. For every formula of this language, at every cut,

$$\neg A \vee \neg\neg A$$

holds.

Proof. In a complete finite record, whether A is realized at `T_E` is determined. If it is, no cut can realize `¬A`; otherwise persistence would give both A and `¬A` at `T_E`. Thus `¬¬A` is realized at every cut. If A is not realized at `T_E`, it is realized at no cut, since otherwise it would persist to `T_E`. In that case `¬A` is realized at every cut. ∎

For an event d of the complete history, we obtain `c_∅ ⊩ ¬¬p_d` but `c_∅ ⊮ p_d`. This distinguishes belonging to the complete history from an already actualized act; it does not introduce a choice for the Universe.

An arbitrary formula must not be substituted for the event d in this claim: the schema `¬¬A` is not a law, since it fails for `A=⊥`.

6.3. The precise meaning of double negation

In a fully specified finite model,

$$c \Vdash \neg\neg A \iff \forall u \geq c\, \exists v \geq u :\ v \Vdash A.$$

This is a statement about continuations of the description. To read it as inevitable actualization along a real local trajectory, the model must also preserve the author's meaning of actual change and completion of a level.

This revision does not introduce a physically halted system as a counterexample to the author's rejection of stasis. A different point is clarified: **the reflexive relation `c≤c` in the language of checks is not a real act without change**. When a mathematical sequence of acts is considered, an elementary step must strictly add an event; for finite E, every maximal such continuation reaches `T_E`. This is a statement about the representation, not the introduction of a global physical tick.

The transition to ontological W itself does not follow from the presence of a greatest element in a set. It requires preservation of the author's meaning of level and self-closure.

6.4. What uniqueness does not mean

For two incomparable events d and e, there are cuts

$$\varnothing, \quad \{d\}, \quad \{e\}, \quad \{d, e\}.$$

Two chains of enumeration pass through `{d}` or through `{e}`. If an order between d and e is not a relation of the original structure, these are two linear representations, not two ontological trajectories. The preceding audit would be inaccurate if it used this example against the author's uniqueness claim.

The reverse inference must not be overstated either: the existence of a common continuation does not by itself prove the author's claim about the uniqueness of reality as a whole. Lemma 6 characterizes a mathematical representation in which one history is already treated as given.

7. Local Sequence and Dummett's Law

7.1. The language of one local system

Let the events of the local system under consideration form a chain `S⊆E` under its own event order. This uses the author's meaning of a local sequence of elementary changes; it does not automatically extend to the union of all systems.

A formula about S contains only atoms for events in S. The projection of a cut into this local record is

$$\pi(c) = c \cap S.$$

Lemma 8. Every local continuation `j⊇π(c)` that is an initial segment of S lifts to the cut

$$u = c \cup \downarrow_E j$$

with `u⊇c` and `π(u)=j`.

Proof. Two down-sets are united. If an event in S precedes an event in j, it belongs to j because j is an initial segment of the induced order on S. The projection therefore adds no extra events to j. ∎

The projection preserves the values of local atoms and all available local continuations. Induction on formulas yields preservation of the meaning of the local connectives `∧,∨,→,¬`.

7.2. Comparability of regions of realization

Theorem 9. For formulas A and B of the local language of S, the following holds at every cut:

$$(A \to B) \vee (B \to A).$$

Proof. On a chain, two persistent sets are comparable by inclusion. Otherwise there are points `x∈U\V` and `y∈V\U`. If `x≤y`, persistence of U gives `y∈U`, a contradiction; if `y≤x`, the analogous contradiction follows for V. Hence either `U⊆V` or `V⊆U`. Accordingly, one of the two dependencies is realized. Lemma 8 transfers the result to the language of cuts of E. ∎

If the chain of cuts has k points, it has k+1 persistent regions: the empty region and k final segments. This is the number of distinct realization profiles of formulas, not the number of ontological outcomes of one event.

7.3. The converse result for the complete event record

Theorem 10. Dummett's law holds on `J(E)` for all formulas of the full event language if and only if E is linearly ordered.

Proof. If E is linear, the cuts also form a chain, and Theorem 9 applies. If d and e are incomparable, take the cut c consisting of all their strict predecessors. The cut `cu{d}` realizes d without e, and `cu{e}` realizes e without d. Thus neither `p_d→p_e` nor `p_e→p_d` is realized at c. Their disjunction is not realized either. ∎

The theorem characterizes the order of events in the model. It does not define a global "now." The existence of incomparable events, the absence of privileged synchronization, and the absence of external time must not be declared synonymous without separate justification.

The corrected interpretation: if incomparable events are established in the history being described, Dummett's law fails for the full event language. It holds for the local language of a chain. Neither result introduces a physical global queue or refutes the uniqueness of consistent actualization.

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8. The Complete Cut, the W of a Level, and Interlevel Relations

8.1. Classical determinacy at the complete cut

Theorem 11. At `T_E`, for every formula of the current event record,

$$A \vee \neg A, \qquad \neg\neg A \to A$$

hold.

Proof. The only continuation of `T_E` in this `J(E)` is `T_E` itself. Hence the exclusion of A coincides with the absence of A at this point, and the conditions for the connectives become two-valued. ∎

By Theorem 5, these schemata do not hold generally at a proper cut. Individual instances may nevertheless hold earlier.

The theorem does not prohibit a classical language from describing change. For example, “property P is absent in state s and present in another state s′” is not a contradiction when the states are distinguished. Nonclassicality in this article pertains to the persistent semantics of realization, not to the impossibility of any other language.

8.2. Why the complete cut is not automatically declared to be W

`T_E` means that all events of the record are included. The author’s `W_q` has a stronger meaning: the realized self-closed whole of a level, with absolute structural indistinguishability of its atoms in the relevant respect, rather than a formal union of independent parts.

Including all elements of a finite list does not by itself entail either the necessary interaction of all parts or their required indistinguishability. The notations are therefore separated not to reject W, but to prevent a substitution:

$$\text{completeness of an event record} \quad \neq \quad \text{an already proved complete description of ontological W.}$$

To apply Theorem 11 specifically to `W_q`, it must be shown that the chosen record preserves the essential relations of that level and that its completeness corresponds to realized closure. This is an obligation of the descriptive language, not a proposal to make W something else.

8.3. Preservation of meaning across a level transition

It does not follow from `qW≡(q+1)0` that all terms and valuations of the old language can be transferred unchanged to the next level. If symbols denote different atoms or different relations, identical notation does not yet ensure identical meaning.

A sufficient mathematical condition for transferring formulas can be stated explicitly. Suppose the new event model has a map π into the previous one that preserves the valuations of the old atoms and the order. In addition, every old continuation must have a corresponding new one:

$$\pi(w) \le u \quad \Rightarrow \quad \exists v \ge w : \ \pi(v) = u.$$

Then, for old formulas,

$$w \Vdash_{new} A \iff \pi(w) \Vdash_{old} A.$$

Proof. The atomic case is given by preservation of meaning. For `∧,∨`, apply induction. For implication, the direction from the old model uses monotonicity of π; in the reverse direction, each old continuation is lifted by the stated condition. Negation is the case of implication into `⊥`. ∎

This is a sufficient bridge between descriptions, not an additional postulate asserting the existence of levels. It is not claimed that this bridge has already been constructed for the entire W-chain.

8.4. There is no “reset of all truth”

The beginning of the next level does not make all formulas undetermined. `⊤`, `A→A`, and the exclusions of impossible conditions retain their meaning. Nonclassicality at a new initial cut is established through the existence of an event of that level that has not yet been included, not through the assertion that “nothing at all is true.”

The earlier text’s “ladder of logics” combined different operations: restricting the language to one subsystem, changing the class of mathematical representations, and moving to a complete record. This is not a proved physical temporal ladder of all statements about the Universe.

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9. The Topology of Regions of Realization

9.1. Topology as a descriptive construction

On 𝒞=J(E), a set U is called open if it is persistent under continuation. For any X⊆𝒞,

$$\operatorname{Int} X = \{c : \uparrow c \subseteq X\}, \qquad \operatorname{Cl} X = \{c : \exists u \geq c,\ u \in X\}.$$

Here ↑c={u:c≤u}. These operations characterize regions of realization in the mathematical record. Identifying them directly with physical space without an additional correspondence would replace the meaning of D2 (the author's definition: space is the topology of distinctions) with something else.

9.2. Exact equalities

Theorem 12. For the connectives constructed in Section 4,

$$\begin{aligned} [A \wedge B] &= [A] \cap [B], \\ [A \vee B] &= [A] \cup [B], \\ [\neg A] &= \operatorname{Int}(\mathcal{C} \setminus [A]), \\ [A \to B] &= \operatorname{Int}\big((\mathcal{C} \setminus [A]) \cup [B]\big), \\ \mathcal{C} \setminus ([A] \cup [\neg A]) &= \partial [A]. \end{aligned}$$

Proof. The first two equalities have already been obtained. The interior of the complement of U consists of cuts all of whose continuations avoid U; this is exclusion. The interior of (𝒞\U)∪V means that there is no continuation with U but without V; this is implication. Finally, [A] is open, so its boundary is Cl[A] \ [A]. Belonging to the closure means that there is a continuation with A, that is, the absence of ¬A. The last equality follows. ∎

Furthermore,

$$\mathcal{C} \setminus ([\neg A] \cup [\neg\neg A]) = \operatorname{Cl}[A] \cap \operatorname{Cl}[\neg A].$$

Thus the weak law of excluded middle corresponds to the absence of an intersection between these two closures.

9.3. What this topology does not assert

The boundary of [A] describes the absence of both the realization of A and its complete exclusion at the given cut. It is not the physical boundary of a body, a proved spatial dimensionality, or a locus of ontological indeterminacy in the Universe.

For the complete cut, {T_E} is open. This does not mean that T_E is separated from its history: on a chain c<T_E, the closure of {T_E} contains both points. Still less does it imply a physical halt or isolation of a W-level.

Identifying this topology of regions of realization with the author's "topology of distinctions" requires preservation of the subject matter's meaning. It is not established merely by the shared word "topology."

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10. Self-Consistency of Reasoning, Self-Reference, and Gödel's Theorems

10.1. Structural closure is not a proof error

Three cases must be distinguished.

Structural self-closure: the relations of the parts constitute a whole in the author's ontological sense. The notation WW≡00 concerns this subject.

Self-application of a description: a language or algorithm considers its own expressions, operations, or executions. This is not in itself an error. Correctness depends on preserving the meanings of terms and justifying the particular steps.

Vicious circularity in justification: a conclusion is used as its own sole ground but is presented as independently obtained. To establish such circularity, one must identify a specific chain of dependencies in the proof. The mere presence of self-reference or a closed diagram is not enough.

In the original article, the relation between the object language and the reasoning tools used is described as a check of self-consistency. This is not the same as a demonstrated derivation of all reasoning tools without any semantic premises. This revision does not claim that a vicious circle has been found in the structural W-chain.

If someone simultaneously writes `T_E=∅`, `T_E=E`, and that E is nonempty in a single event model, the resulting description is contradictory. But this is not the literal meaning of the author's `WW≡00`: it does not identify an empty list of events with a nonempty one within the same vocabulary. The error in such a representation does not transfer to ontological self-closure.

10.2. Why reasoning does not prescribe laws to the Universe

A mathematical proof is itself an activity of a subsystem and is expressed in a language. It must preserve its subject matter, the meaning of its statements, and the necessary dependencies between them. This does not make the notation used into the ontological source of the structure being investigated.

In Sections 4–9, the meanings of the connectives are first explicated through relations of realization, and their consequences are then obtained. Finite sets, inclusion, and reasoning by the structure of a formula are used to describe these relations. The use of this apparatus is stated explicitly. It is not presented as an external law that the Universe is obliged to obey.

Two equally important limitations follow:

One must not reject an ontological ground merely because its exposition uses logical means: the origin of a ground and the way it is expressed are different questions.

One must not declare every step correct merely because every actual instance of reasoning occurs in a self-consistent Universe. The actual existence of an erroneous inscription does not make its content true. References to `C=0` therefore do not replace checking an inference.

10.3. What the incompleteness theorems actually assert

The subject here is the **incompleteness theorems**, not every result of Gödel's and not every object that can be called a "system."

The first theorem, in the standard modern Gödel–Rosser form, concerns a consistent, effectively axiomatized theory capable of expressing sufficient arithmetic, for example one containing Robinson arithmetic Q. For such a theory T, there is a sentence φ for which

$$T \nvdash \varphi, \qquad T \nvdash \neg\varphi.$$

Gödel's original version used the stronger condition of ω-consistency; Rosser's strengthening permits ordinary consistency in the specified class of theories. [G1, G2]

The second theorem concerns a formalized statement of the theory's own consistency. With sufficient arithmetical strength and the standard representation of provability, a consistent T does not prove its own `Con(T)`. The precise meaning of `Con(T)` and the properties of the formal provability predicate are essential. This is not a prohibition on every verification of every software object, nor a claim that all self-reference is erroneous. [G1]

These results are neither refuted here nor invoked as an appeal to authority against W-structure.

10.4. In what sense ontological W-structure lies outside their direct subject matter

Let $\mathscr{W}$ denote ontological actualization, and T a particular formal theory describing it. In the author's conception, $\mathscr{W}$ is not automatically assigned a list of formulas, recursively enumerable axioms, and an internal relation `T⊢φ`. To apply a theorem about T, that particular kind of object must first be exhibited and the theorem's premises checked.

It is therefore correct to say:

> **Ontological self-closure W is not the formal completeness of a calculus. The incompleteness theorems do not establish the impossibility or inconsistency of a self-closed Universe merely because it is called a system.**

This distinguishes subject matters; it is not a “victory over Gödel.” The word “above” can indicate only the primacy of ontology in the author’s conception, not a proved immunity of every language describing W to the limitations of formal theories.

The converse limitation is equally important: the self-consistency of $\mathscr{W}$ does not imply that every subsystem possesses a complete algorithmic description of $\mathscr{W}$. If a particular theory T meets the conditions of the incompleteness theorems, those conditions do not disappear when its axioms are renamed “necessary relations.” And if W is not a theory of that kind, the theorems’ inapplicability does not prove all claims about W.

10.5. Three different kinds of completeness

Concept	Subject matter
Ontological self-closure	The relations of an existing whole in the W conception
Semantic determinacy	The truth of a statement under a fixed meaning and a complete mathematical structure
Syntactic completeness	The ability to prove or refute every sentence using a particular calculus

A result must not be transferred from one row to another without a proved correspondence. What is unknown to a subsystem does not become an undecided Universe. Incompleteness of a calculus does not imply ontological contradiction. Nor is ontological self-consistency the same as a calculus’s ability to prove its own consistency.

This distinction is compatible with the content of Gödel’s theorems themselves: limitations on provability are not identified with the absence of a genuine subject matter for the description. [G1, G3]

10.6. Finiteness must not be used as a merely verbal escape

In a fully specified finite event record, the value of a particular finite formula can be computed by a finite case analysis. For an explicitly given finite structure with a finite computable vocabulary, its first-order sentences can likewise be checked through finite quantification. These are precise statements about fixed objects.

But “every expression and proof is finite” does not mean “there is a single finite list of all admissible expressions and proofs.” The incompleteness theorems concern precisely finite inscriptions whose length has no fixed upper bound and that have arithmetical expressive power. The author’s rejection of ontological infinity does not by itself establish which mathematical premise fails for a particular descriptive calculus. [G1, G2]

Likewise, successive extension of a proof program is not an automatic escape: if the union of its rules is effectively enumerable, consistent, and retains the requisite arithmetical strength, the corresponding conditions apply to the resulting formal apparatus. This follows from checking the premises; it is not an objection to the development of VM + S (the author’s local proof-and-verification system).

Conclusion of this section: ontology must not be replaced by a formal language; a formal language must not be granted privileges unsupported by its actual properties.

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11. Exact Mathematical Correspondences and Reproducible Verification

11.1. Which properties of the representation characterize the schemata

Let a finite partial order $\mathscr{C}$ be given, and let variable valuations range over **all** persistent regions. Then, at a point c:

Schema under all persistent valuations	Necessary and sufficient condition
`¬A∨¬¬A`	Any two continuations of c have a common continuation
`(A→B)∨(B→A)`	The cone of continuations of c is linearly ordered

Schema under all persistent valuations	Necessary and sufficient condition
`Av¬A`	c has no proper continuation
`¬¬A→A`	c has no proper continuation

Justification of the converse directions. If u and v have no common continuation, put `[p]=↑u`. At c, `¬p` is not true, because p is realized at u. But `¬p` is realized at v, so `¬¬p` is not true at c.

If u and v are incomparable, put `[p]=↑u`, `[q]=↑v`. At u, p is realized without q; at v, q is realized without p. Dummett's law fails at c.

If there is a `u>c`, the valuation `[p]=↑u` refutes excluded middle at c. For double negation, `[p]=↑c\{c}` suffices: p is realized at every proper continuation, `¬¬p` is realized at c, but p is not realized at c.

Justification of the forward directions. For the first property, the absence of A from every continuation gives `¬A`. If A is realized at some u, a common continuation of u and each v rules out `¬A` at v, and therefore gives `¬¬A` at c. Linearity makes persistent regions comparable, as in Theorem 9. Terminality reduces the semantics to a single point, as in Theorem 11. ∎

The qualification about all valuations is essential. On a nonterminal chain, one could permit only constant atom valuations $\varnothing$ and $\mathscr{C}$. All resulting formulas would have constant two-valued valuations. Such a restricted language cannot establish the absence of continuations.

In the complete event record, the expressive power required for these converse inferences is provided by the formula `Φ_U` in Section 3.4.

11.2. Forms without disjunction

From `PvQ` follows

$$(P \to C) \to ((Q \to C) \to C)$$

for every C. Conversely, **the entire schema over C** yields `PvQ` by substituting `C=PvQ`. A single instance with an arbitrary fixed C, however, is not equivalent to `PvQ`.

For example, when `C=⊤`, the implicational formula is true regardless of P and Q. On a chain `0<⊤` with p only at T, excluded middle `p∨¬p` is not true at 0. Thus the formula with `C=⊤` cannot be presented as equivalent to excluded middle at that point.

This correction also applies to the earlier forms of the weak law of excluded middle and Dummett's law.

11.3. A different "or" denotes a different relation

One can define

$$A \vee_w B := \neg(\neg A \wedge \neg B).$$

In the semantics under consideration, this is equivalent to `¬¬(A∨B)`, not to the original disjunction. It means the absence of joint exclusion, but not yet the realization of one of the disjuncts at the given cut.

On the chain `0<⊤` with `[p]=[r]={⊤}`, the statements `p∨_w¬p`, `p→r`, and `¬p→r` are realized at 0, but r is not. Ordinary reasoning by cases therefore does not work for `∨_w`. Choosing another word or symbol does not change the Universe; it changes the condition being expressed.

11.4. Names of known logics and the precise scope of their application

The meanings of the connectives in the resulting language coincide with the intuitionistic semantics of persistent propositions. IPC is the name of a known calculus that is sound for such models. It is not an additional ontological ground of this article.

For the class of **all** finite event records `J(E)`, under all persistent valuations, the exact set of valid schemata is KC, the logic of the weak law of excluded middle.

$$\boxed{\mathrm{Log}\{J(E) : E \text{ is finite}\} = \mathrm{KC}.}$$

The justification consists of two inclusions. Each such frame has a greatest element, so KC is sound on it. If E is an antichain of n events, `J(E)` is the full Boolean cube `2^n`, ordered by inclusion. For these cubes, it has been proved that

$$\mathrm{KC} = \bigcap_{n \geq 1} \mathrm{Log}(2^n).$$

The latter is an external mathematical lemma [L1, Corollary 1, p. 1052], not a derivation from the author's ontology. All these cubes belong to the class under consideration; this gives the reverse inclusion. ∎

For one fixed finite history, the set of schemata may be stronger. "The logic of the entire class of representations" does not mean that the Universe actualizes all these mathematical representations. In particular, without a separate bridge, the equality with KC is not declared to be the definitive classification of the entire ontological logic of W.

11.5. Verification performed for revision 4

A new program, `verify_revision4.py`, was written and run for this revision. It checks operations on finite descriptive frames, not the author's ontology. Neither the historical programs from the original article nor the local VM + S system was run.

Check	Actual scope
All labeled partial orders on 1–5 points	4,473
Persistent valuations of one variable	48,710
Persistent valuations of two variables	573,836
Triples for the characterization of implication and two composition laws	7,316,312
Pointwise checks of four frame correspondences	88,380
Lattices of cuts of orders on 1–4 events	242
Projections onto nonempty local chains	2,040
Persistent regions on cuts checked, including expressibility	3,709
Comparisons of the topological and order-theoretic representations of implication on cuts	105,249

All program assertions passed. The code (`verify_revision4.py`) and its actual output (`verification_results.json`) are included in this repository. General proofs are given in the text; finite enumeration does not replace them. Coq, `coqchk`, and VM + S were not run in this work.

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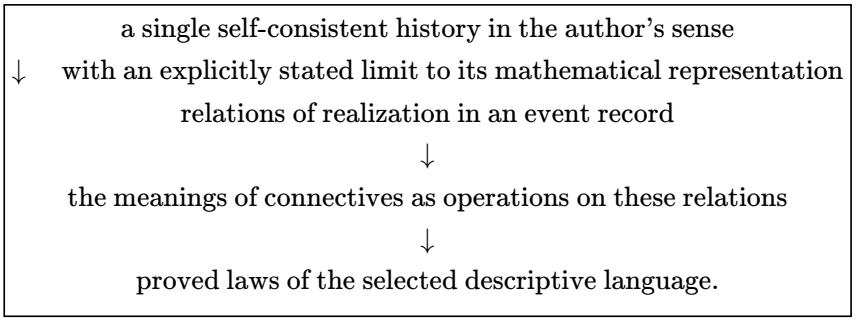
12. Summary: Corrections to the Direction of Derivation

12.1. The article's central claim

In the author's adopted sense, the Universe does not become logical after states have been selected. It cannot exist non-self-consistently. Logic expresses the necessary relations of such existence.

The mathematical description in this article follows that direction: it does not prescribe ready-made axioms to reality, but assigns precise meanings to expressions and checks which relations are preserved when they are used.

The result is:



The first step is not a declaration that any partial order is already a complete description of the Universe. The remaining mathematical steps have been stated and proved.

12.2. What withstood rigorous examination

The following are retained: persistence of historical statements, the law of non-contradiction, semantic explosion, the listed laws of implication, the weak law of excluded middle on cuts of one complete record, Dummett's law for the local language of a chain, classical determinacy at the complete cut, and the topological expression of what is unrealized but not excluded.

An explicit proof has been added that implication and negation are uniquely determined as the greatest persistent regions with the corresponding meaning of compatibility. This makes the content of the former F3 explicit (in revision 2, F3 was the principle that a statement holds at a configuration exactly when it can no longer cease to hold); it does not declare F3 to be a consequence of a single unexplained symbol.

The object of the criticism of uniqueness has been corrected: different orders of recording independent events are not a counterexample to the author's unique trajectory. Self-closure `WW≡00` is not equated with the error of circular justification.

12.3. Which representational steps are not presented as proved

Representational step	Precise status
From ontological self-consistency to the complete apparatus of a particular language	The meanings of the connectives are explicated; no claim is made that there is only one possible language
From an actual history to all cuts `J(E)` as its complete description	All essential relations must be preserved, not merely precedence
From a local discrete tick to the impossibility of a global "now"	The derived character of local time is retained; discreteness alone does not supply such a proof
From `T_E` to ontological `W_q`	A bridge is needed between completeness of the record and realized self-closure
From persistence within a level to the transfer of all old formulas between levels	A sufficient condition on a map is given; its realization for the entire chain is not shown
From the subject matter of Gödel's theorems to the ontological Universe	Automatic identification is inadmissible
From "W is not a formal calculus" to the completeness of every description of W	This does not follow; applicability conditions must be checked for each formal description

This is not a list of new axioms of the Universe. It marks the limits of this particular text's responsibility: where a proof of descriptive correspondence has been supplied and where it has not.

12.4. Finite quantifiers and physical interpretation

For an explicitly specified finite constant domain of objects, quantifiers can be expanded into finite conjunctions and disjunctions. For example,

$$\forall x\,(A \vee B(x)) \to (A \vee \forall x\,B(x))$$

is valid in this semantics if x does not occur free in A. A varying domain raises a separate question. The propositional part of the article does not replace the study of predicates and arithmetic.

Nor does it automatically yield three-dimensionality, a physical metric of space, quantitative clock rates, or particle properties. Preserving the author's ontological meaning does not authorize adding unproved physical consequences to mathematical theorems.

12.5. Conclusion

The Universe does not follow logic as an external command. In the author's conception, logic expresses the relations of a Universe that cannot exist non-self-consistently.

The article does not turn this idea into an initial list of axioms of a selected calculus. It shows how to construct a verifiable language of historical realization, which laws arise in it, and where its connection to ontology still requires a precise formulation.

The self-consistency of the whole, the existence of a finite proof, and a subsystem's knowledge remain distinct subjects. Maintaining this distinction preserves both the intent of W-structure and the precision of the mathematical results — without appealing to the standing of any scientific school and without declaring the unproved to be proved.

Sources and Provenance of Claims

[S0] Original article. "Logic as a Consequence of the Non-Contradiction of the Universe," revision 2, 1 November 2026 (unpublished). The author's definition D2 and the earlier formalization F3 mentioned above are taken from that text.

[S1] Author's clarifications of 9 November 2026. Self-consistency is neither achieved nor selected; one trajectory does not mean a global queue; a nonzero elementary act gives rise to its own nonzero local tick; a point is a nonzero indivisible atom; mathematics is a language, not ontology; structural closure is not equated with circular proof.

[S2] The previous revision 3 and the ChatGPT audit. These were used as objects of renewed examination. Interpretative errors from the previous audit have not been carried into the new text as grounds for inference.

[L1] L. L. Maksimova, D. P. Skvortsov, V. B. Shekhtman. "The Impossibility of a Finite Axiomatization of Medvedev's Logic of Finite Problems." Doklady Akademii Nauk SSSR, 245:5 (1979), 1051–1054. Corollary 1, p. 1052, on the logic of full Boolean cubes. Math-Net: dan42643. The text and page images for pp. 1051–1052 were rechecked. Used only as an external mathematical lemma in Section 11.4.

[G1] Kurt Gödel. "Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I." Monatshefte für Mathematik und Physik 38 (1931), 173–198. DOI: 10.1007/BF01700692. Propositions VI and XI were checked against B. Meltzer's English translation, *On Formally Undecidable Propositions of Principia Mathematica and Related Systems* (1962), printed pp. 57 and 70; the page images were inspected. Used to specify the exact scope of the theorems, not as an ontological premise.

[G2] Barkley Rosser. "Extensions of some theorems of Gödel and Church." Journal of Symbolic Logic 1:3 (1936), 87–91. DOI: 10.2307/2269028. The author's formulation of the strengthening and its conditions were checked against the published extract of the article.

[G3] Russell O'Connor. *Essential Incompleteness of Arithmetic Verified by Coq* (2005), arXiv:cs/0505034. A primary report on a formal proof of the Gödel–Rosser theorem. Used here to distinguish an internal formal theory from external verification; its code was not run in this audit.

The mathematical sources are not presented as established ontology.

Prepared with the assistance of AI models (Claude, ChatGPT).